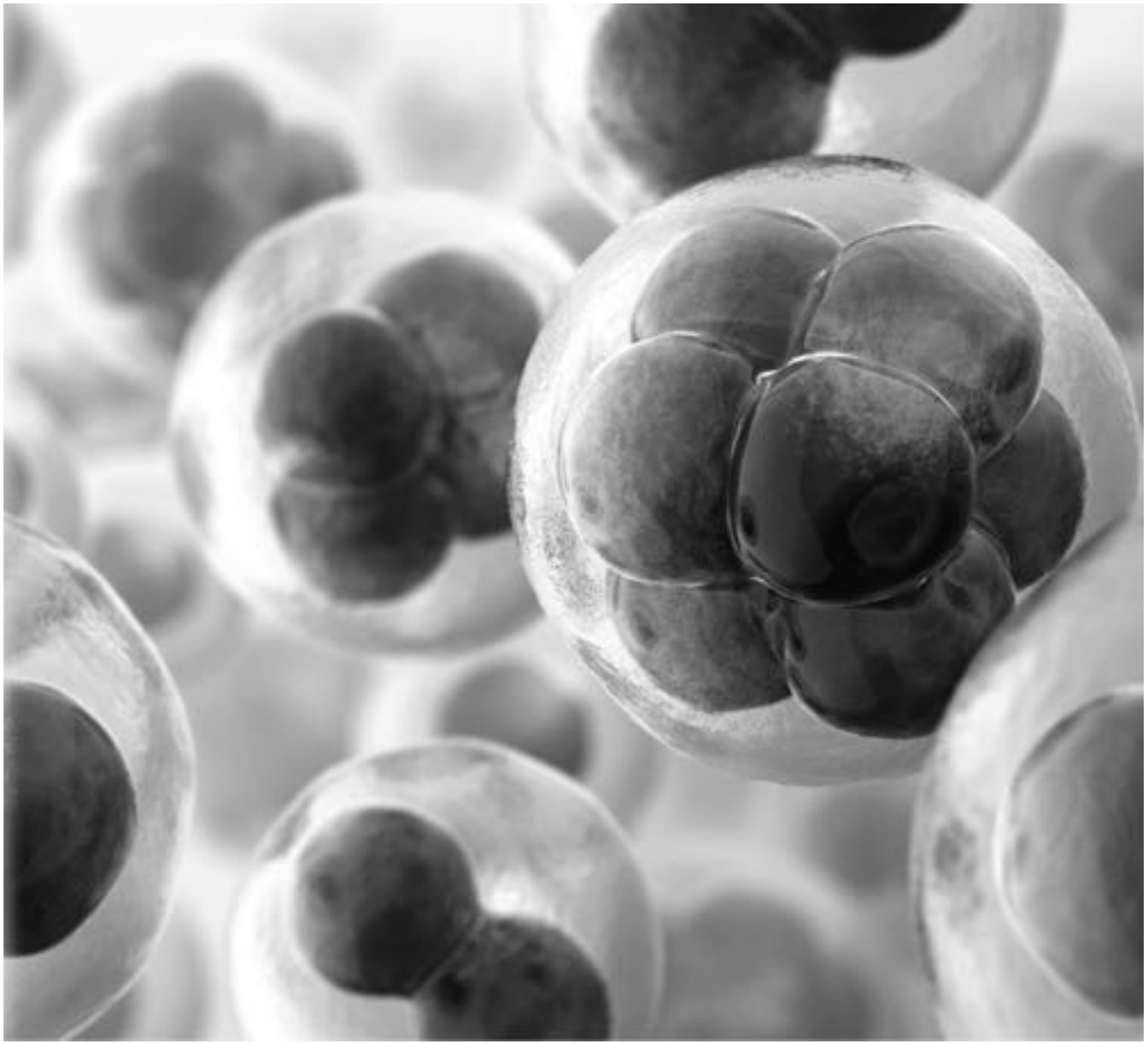
Introduction to Embryology

Embryology

The study of the origin and development of an organism.



Embryology Sequence of Events:

1. Fertilization.
2. Cleavage.
3. Gastrulation.
4. Organogenesis.

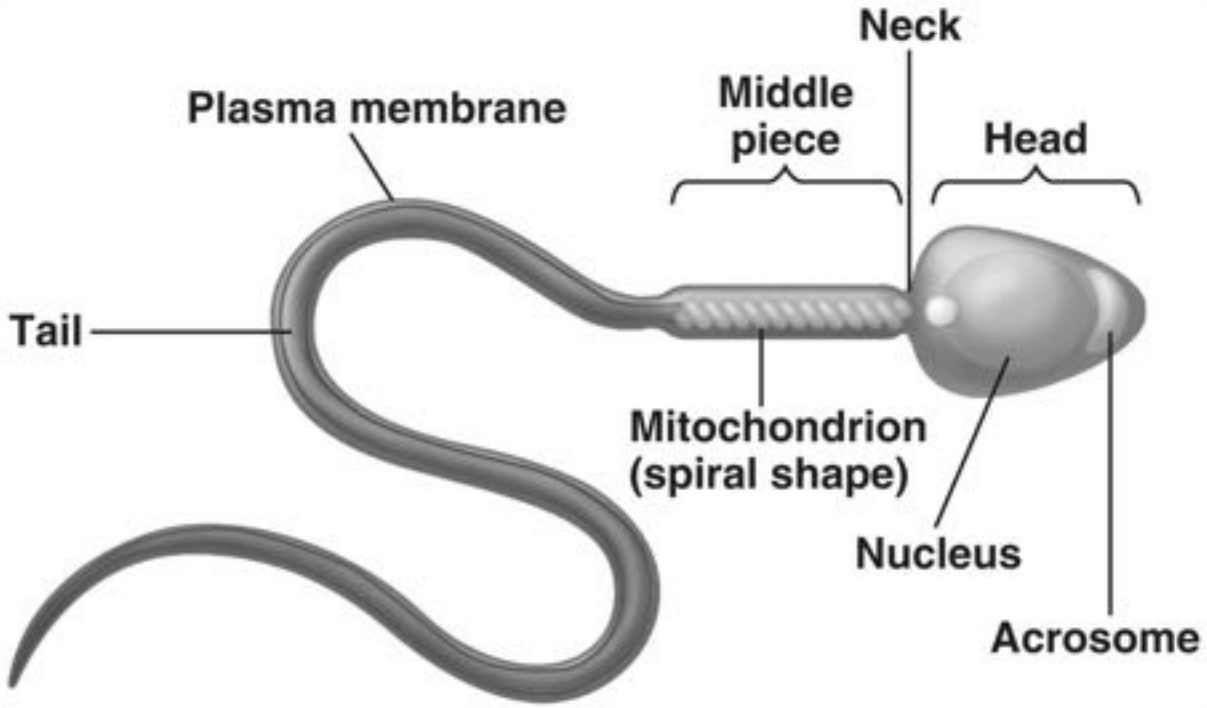
Gametes

In human, sperm 22 somatic chromosomes + 1 sex chromosome (X or Y).

In human, ovum 22 somatic chromosomes + 1 sex chromosome (X).

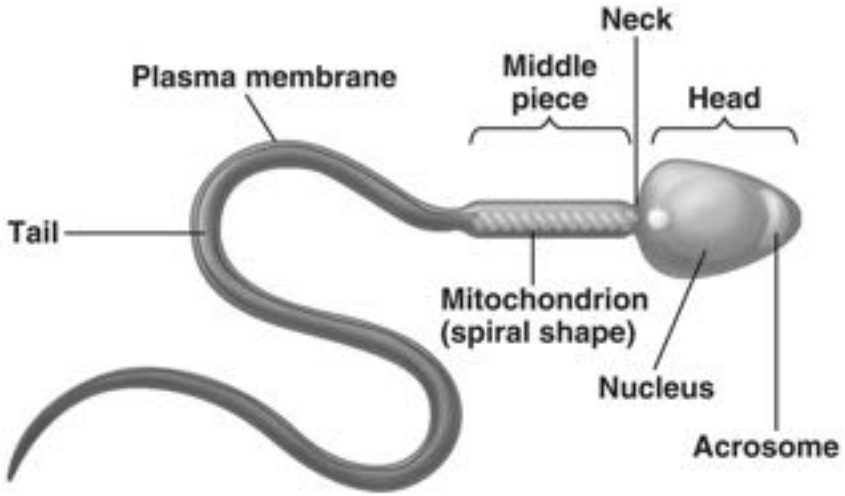
After fertilization, zygote is either contain 44 somatic chromosome + (XX or XY).

Sperm



- **Consists of Head, Middle piece and Tail.**

Sperm



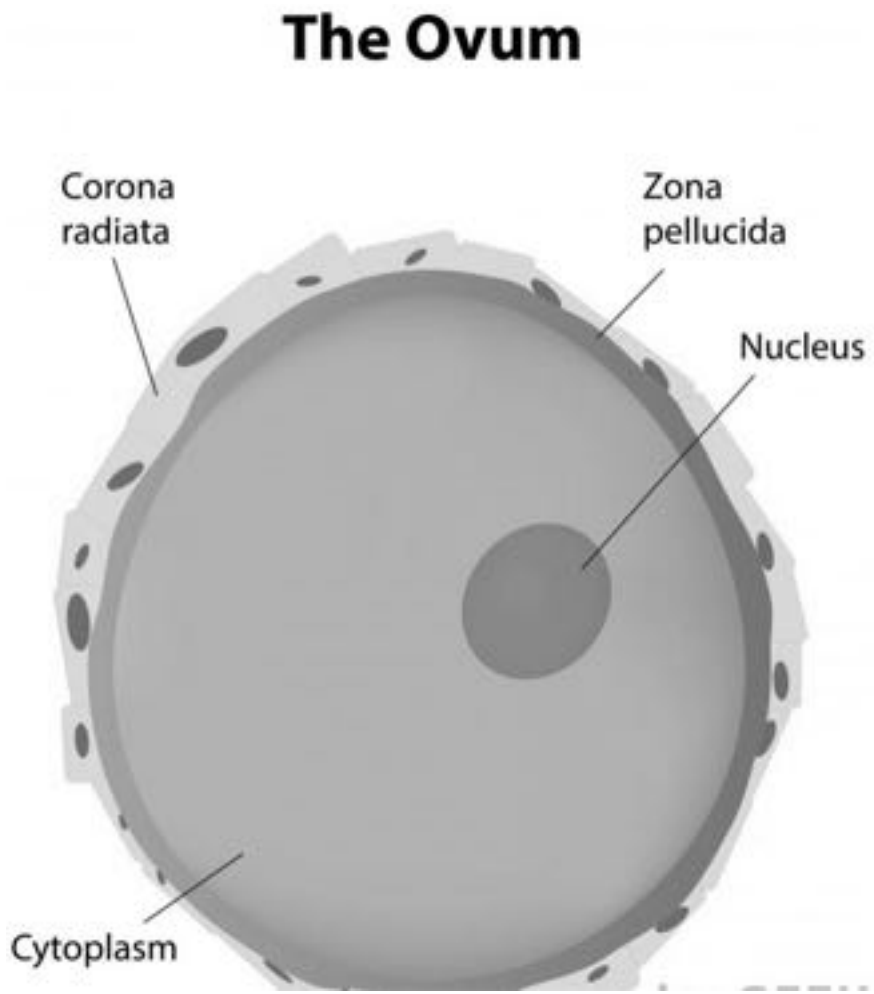
Head: includes the nucleus + haploid chromosomes, including lysosomes with lytic enzymes and cytoplasm.

Middle piece: Centrioles + Mitochondria (Provide energy for tail movement).

Tail: Made of microtubules and microfilaments.

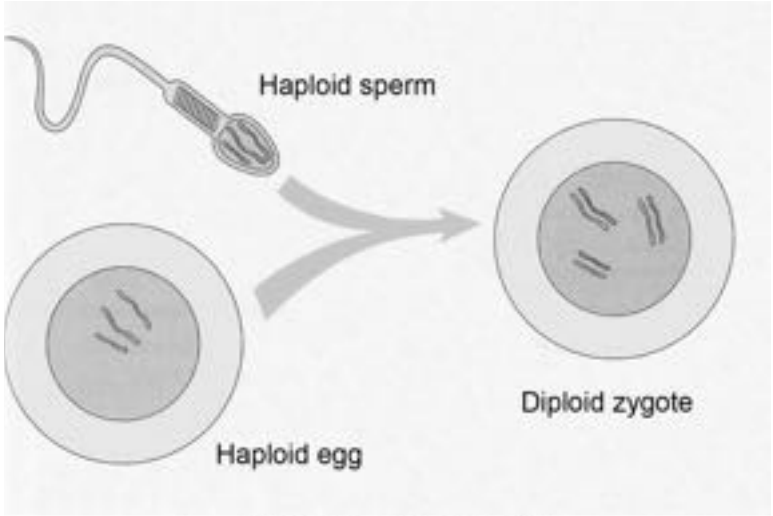
Ovum

- **Consists of a nucleus with haploid no. of chromosomes surrounded by a clear area and cytoplasm.**

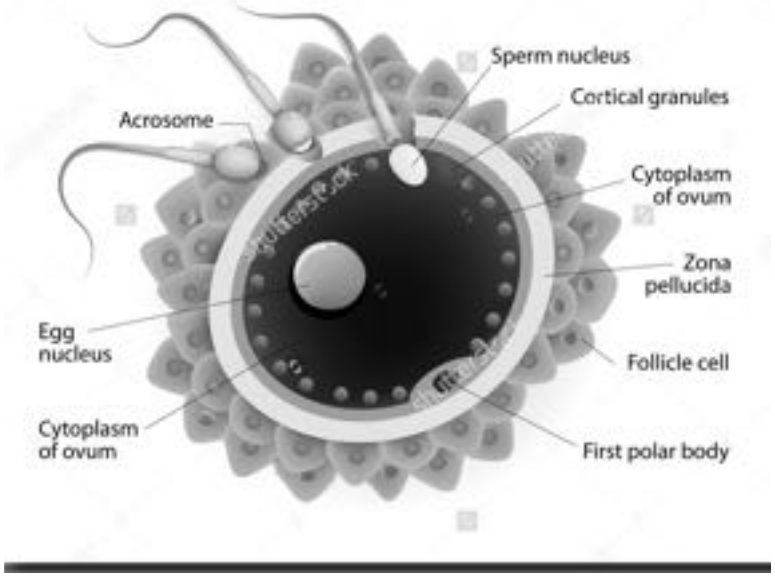


Fertilization

- **Sperm fusion with an ovum leads to embryo development within the body of the female.**

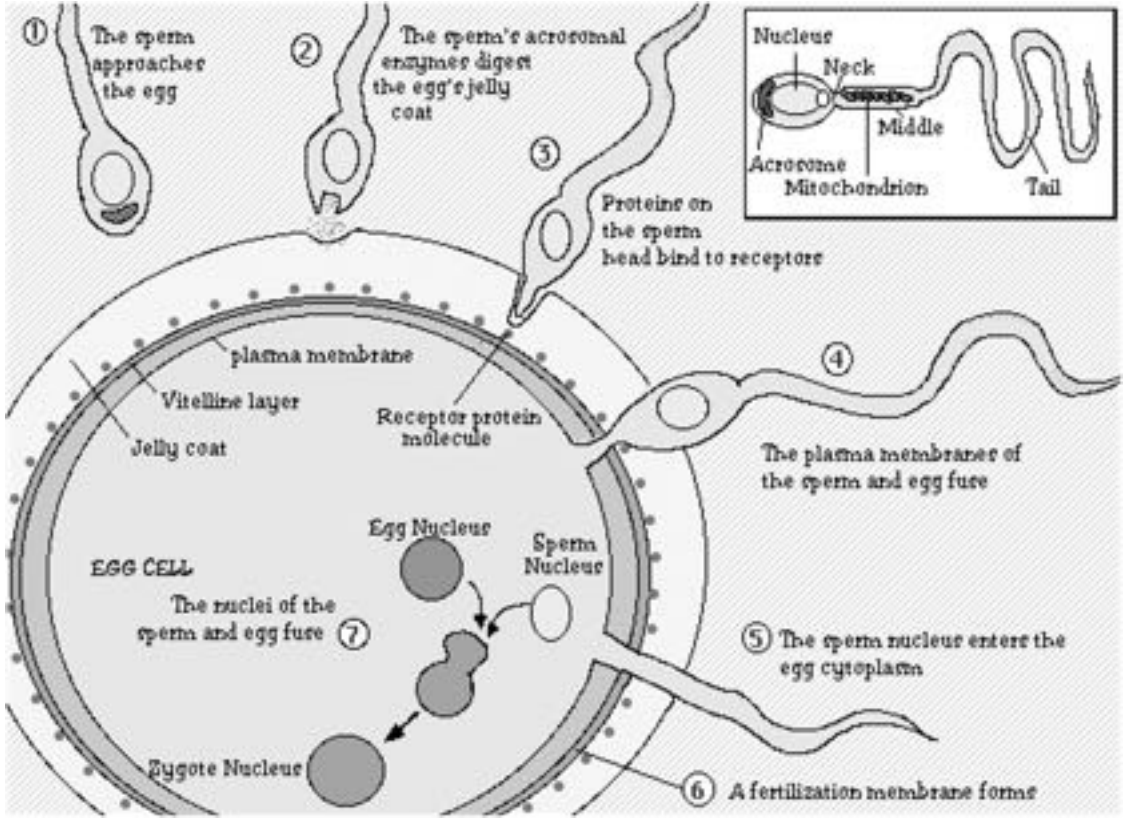


FERTILIZATION



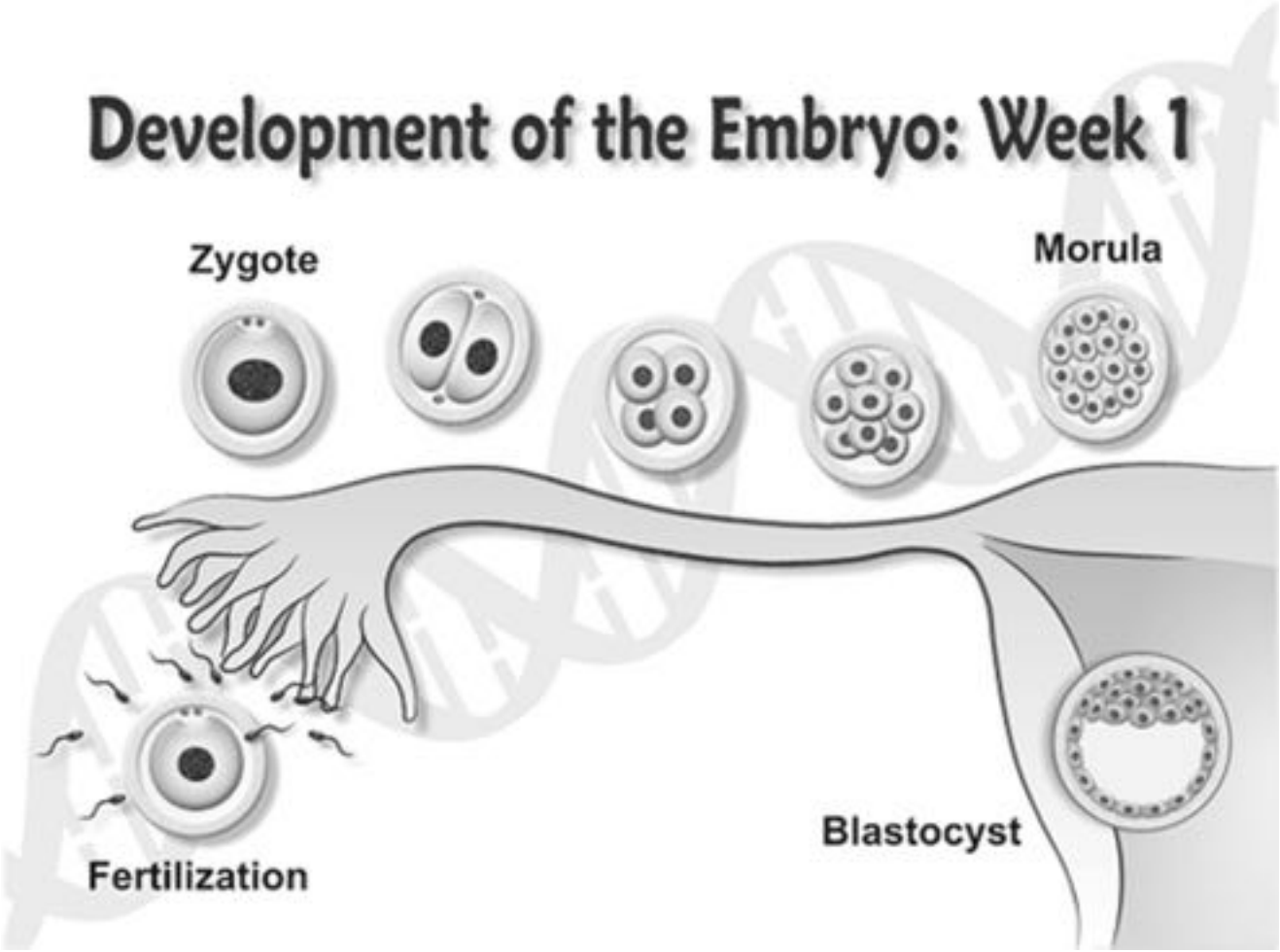
Steps in Fertilization

- Contact between sperm and egg.
- Entry of sperm into egg.
- Fusion of sperm and egg nuclei.
- Activation of development.

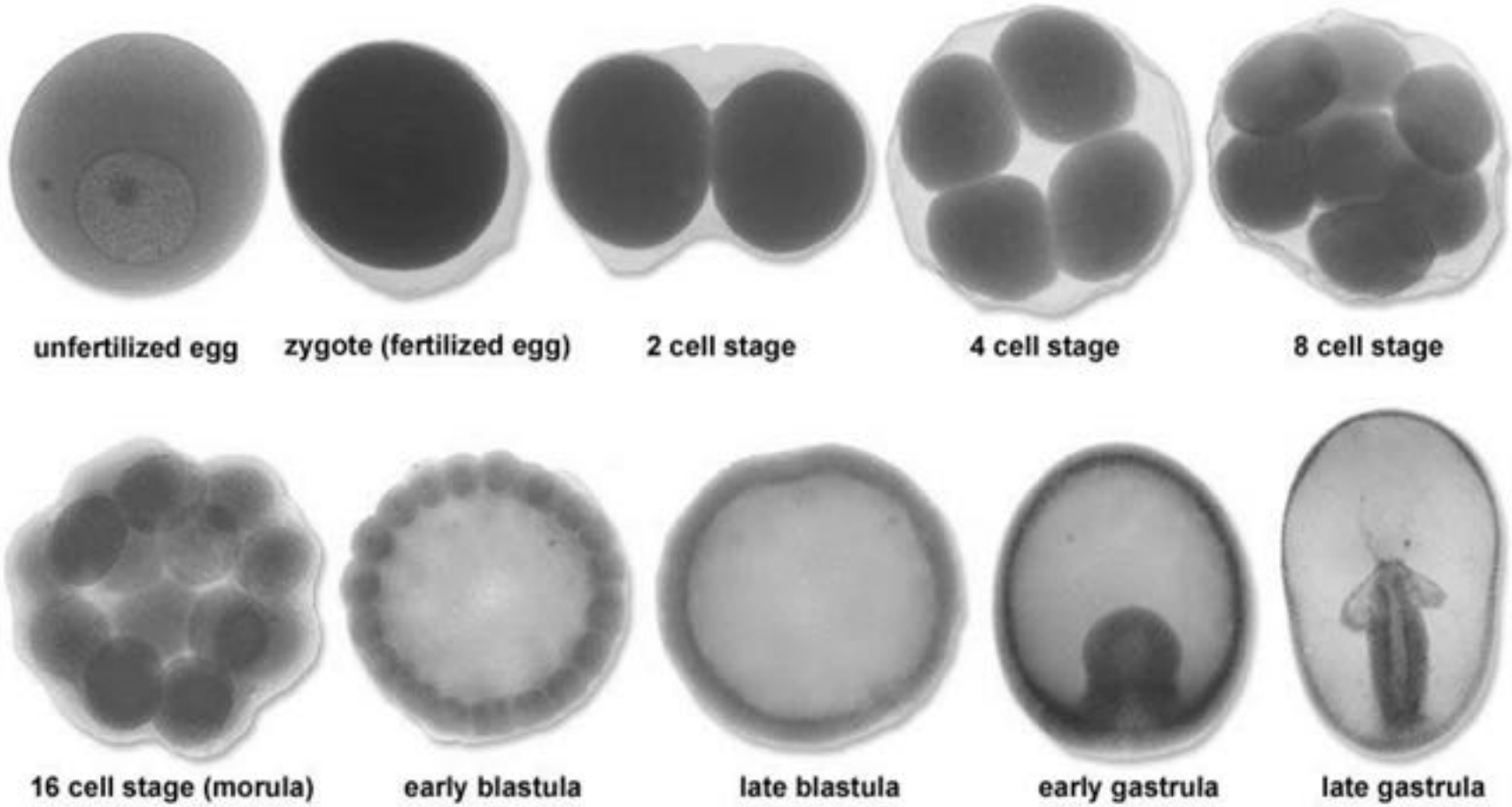


Zygote

- **Diploid cell results from sperm and ovum fertilization.**

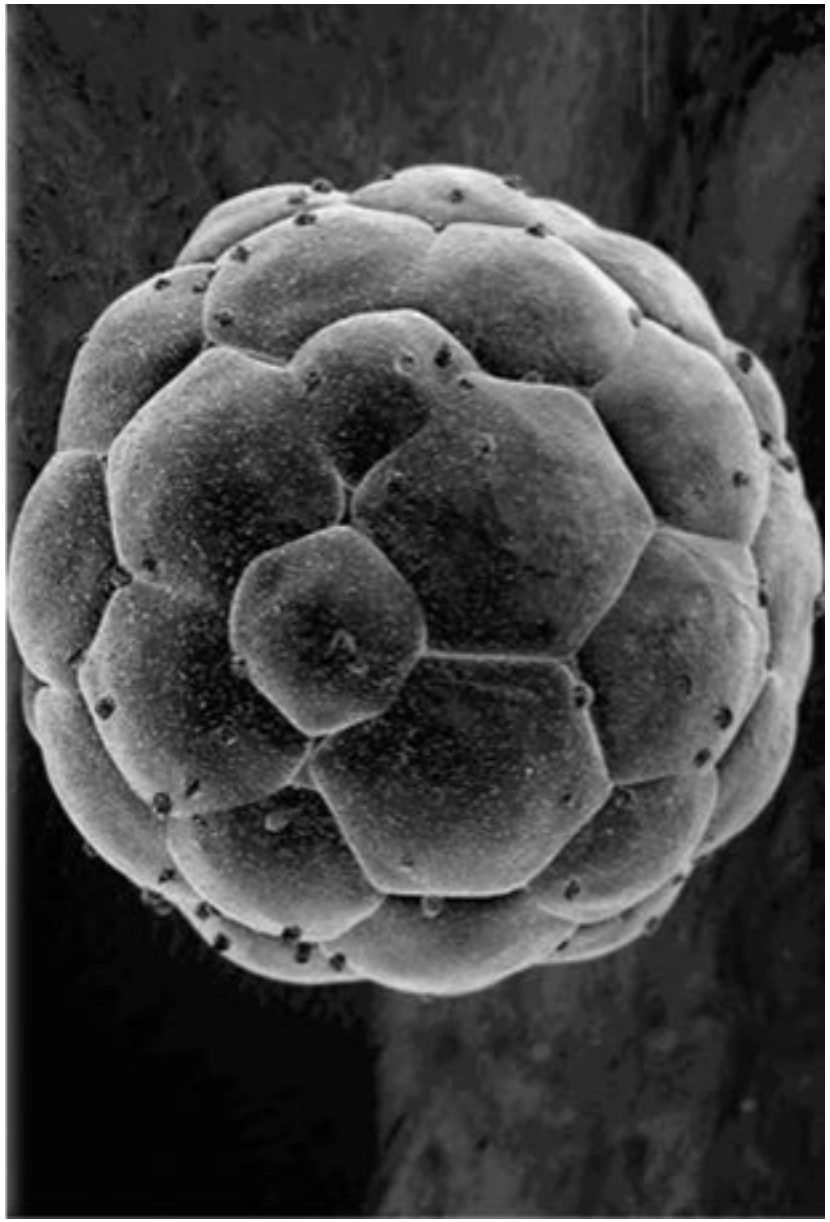

Development of the Embryo: Week 1
Zygote
Morula
Fertilization
Blastocyst

Cleavage


unfertilized egg
zygote (fertilized egg)
2 cell stage
4 cell stage
8 cell stage
16 cell stage (morula)
early blastula
late blastula
early gastrula
late gastrula

Cleavage

- ❖<u>Cleavage</u> is a series of rapid mitotic divisions (without cell growth)
- ❖The two-celled zygote divides repeatedly until a ball of 32 cells is formed
- ❖This is the <u>morula</u> - 32 cells →



Cleavage

Gastrulation = formation of a gut.

- **At the end of the cleavage stage, cells making up the blastula move about and surface proteins help cells recognize each other.**
- **Location of cells after GASTRULATION determines further development.**

Organogenesis

Organogenesis Purpose:

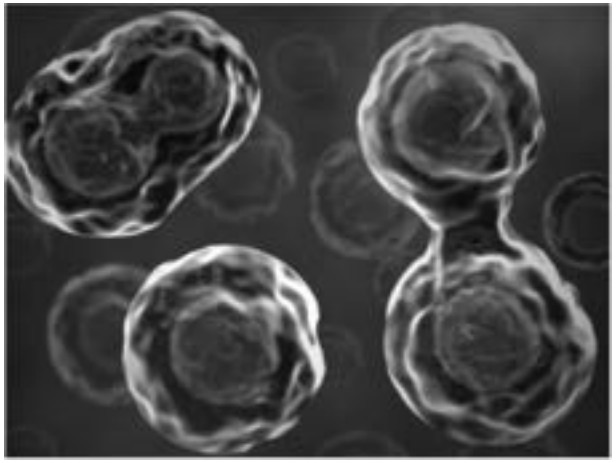
- ❖Formation of organs from three germ layers
- ❖Differentiation & continued Morphogenesis

1. **Endoderm will form the following:**
   - **Epidermis**
   - **Lining of mouth & rectum**
   - **Cornea of eye**
   - **Lens of eye**
   - **Nervous system**
   - **Thin linings of gut & branches**
   - **lining of excretory ducts, bladder**
   - **Lining of lungs, trachea**
   - **Lining of reproductive ducts, uterus, vas deferens**
   - **liver**
   - **pancreas**

Organogenesis

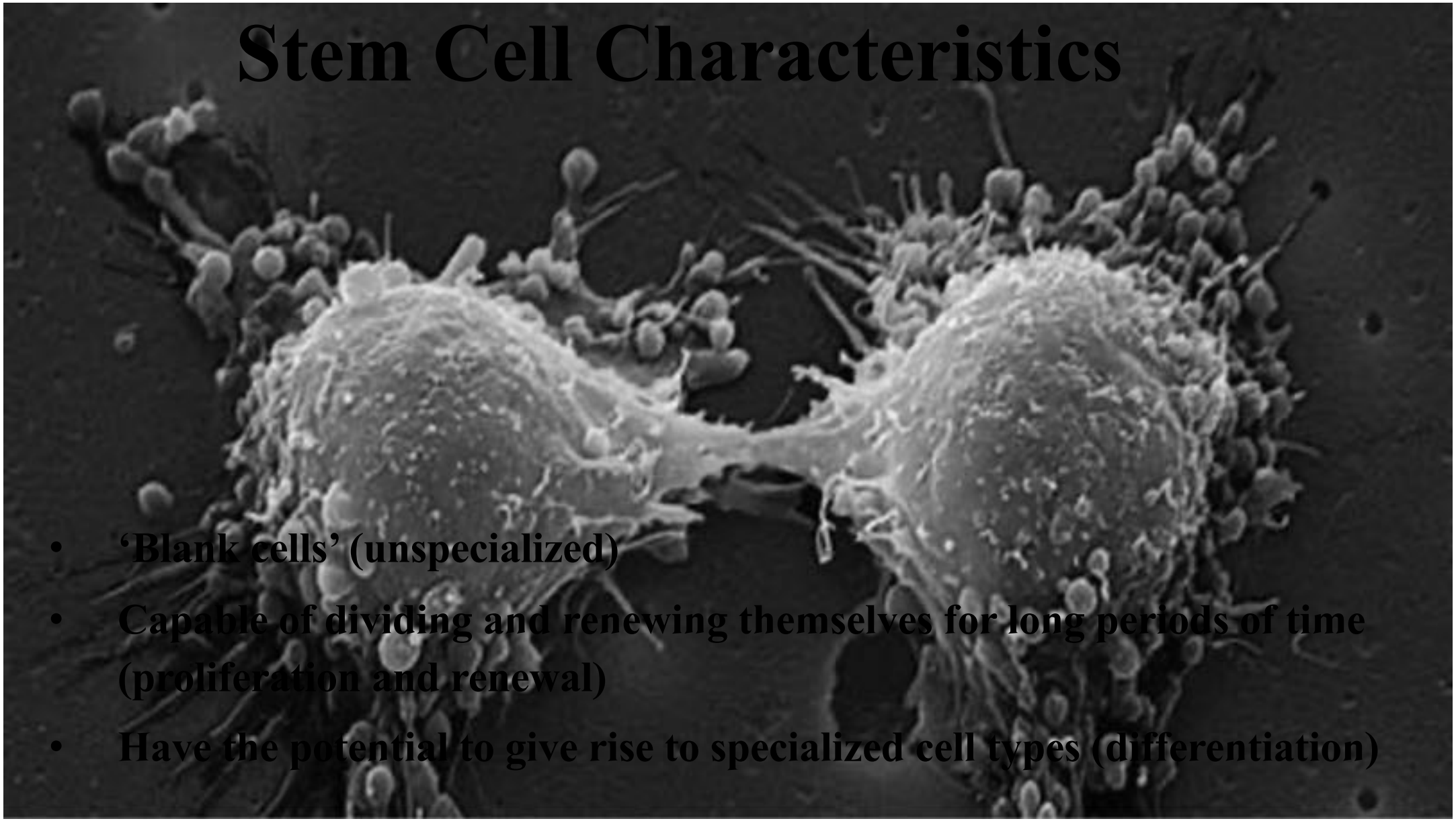
2. **Mesoderm will form the following:**
   - **Skeleton,**
   - **Muscles (skeletal, smooth, cardiac)**
   - **Dermis of skin**
   - **Heart, blood, blood vessels**
   - **Kidneys,**
   - **Ovaries/testes, etc.**
3. **Ectoderm forms the following:**
   - **Skin**

Stem Cell – Definition



- **A cell that has the ability to continuously divide and differentiate (develop) into various other kind(s) of cells/tissues.**

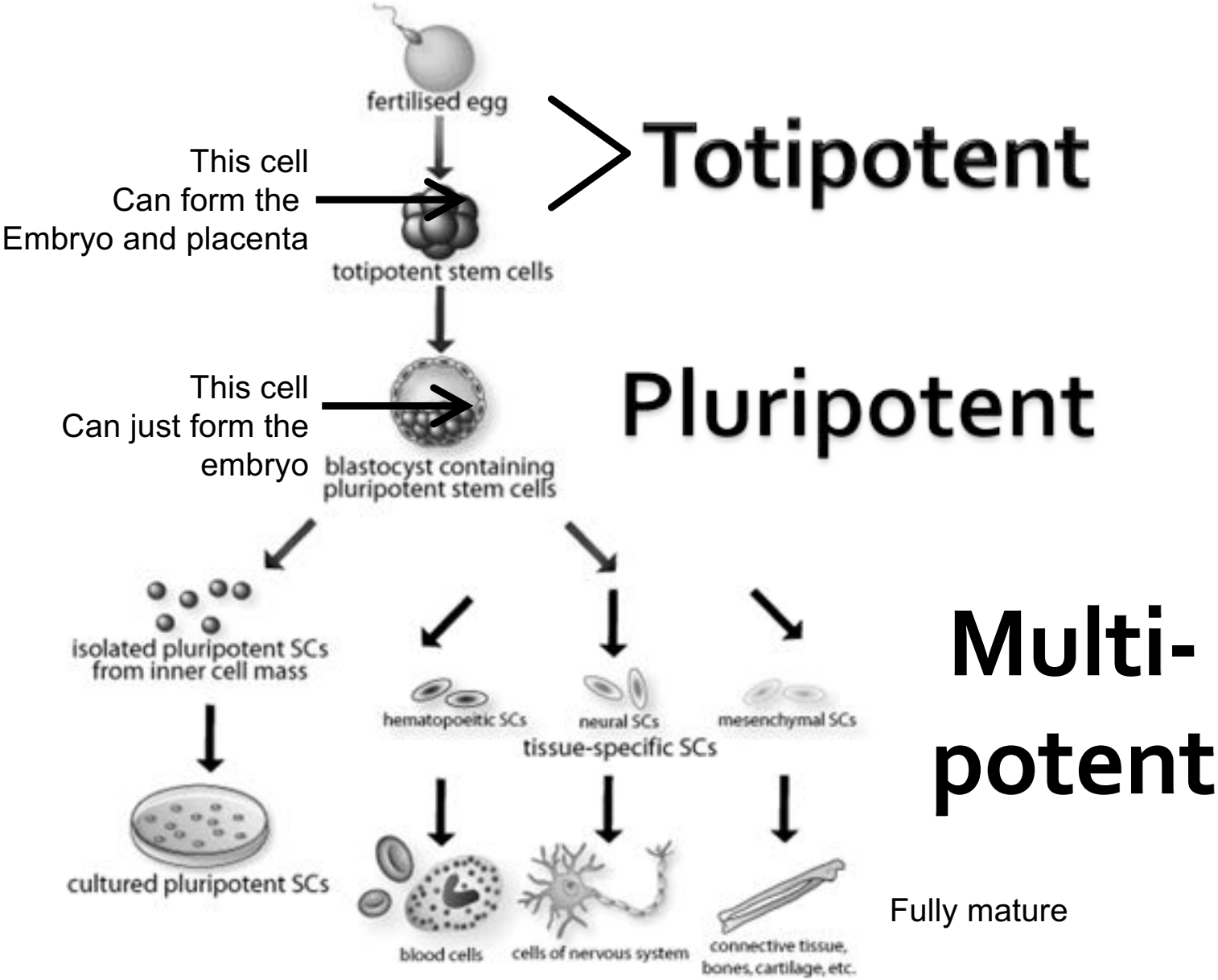
Stem Cell Characteristics



- ‘Blank cells’ (unspecialized)
- Capable of dividing and renewing themselves for long periods of time (proliferation and renewal)
- Have the potential to give rise to specialized cell types (differentiation)

Kinds of Stem Cells

Stem cell type	Description	Examples
Totipotent	Each cell can develop into a new individual	Cells from early (1-3 days) embryos
Pluripotent	Cells can form any (over 200) cell types	Some cells of blastocyst (5 to 14 days)
Multipotent	Cells differentiated, but can form a number of other tissues	Fetal tissue, cord blood, and adult stem cells


fertilised egg
This cell
Can form the
Embryo and placenta
totipotent stem cells
Totipotent
This cell
Can just form the
embryo
blastocyst containing
pluripotent stem cells
Pluripotent
isolated pluripotent SCs
from inner cell mass
cultured pluripotent SCs
hematopoeitic SCs
neural SCs
mesenchymal SCs
tissue-specific SCs
blood cells
cells of nervous system
connective tissue,
bones, cartilage, etc.
Multi-
potent
Fully mature

Kinds of Stem Cells

Embryonic stem cells

- **five to six-day-old embryo**

Embryonic germ cells

- **derived from the part of a human embryo or fetus that will ultimately produce eggs or sperm (gametes).**

Adult stem cells

- **undifferentiated cells found among specialized or differentiated cells in a tissue or organ after birth**
- **appear to have a more restricted ability to produce different cell types and to self-renew.**

Placenta

- **The boundary between the mother's and embryo's system. The umbilical cord extends from placenta to embryo and transport food/wastes.**

Trimester	Month	Week
first	one	1-4
	two	5-8
	three	9-13
second	four	14-17
	five	18-21
	six	22-26
third	seven	27-30
	eight	31-35
	nine	36-40

1st Trimester

- The 3 embryonic layers are formed.
- Cellular differentiation begins to form organs.
- After one month, embryo is 5mm.
- 2nd month, limb buds develop.
- 7th weak embryo becomes fetus.
- 8th weak sex determination starts.

2nd Trimester

- **The fetus increases in size.**
- **Bones starts to form.**
- **Fetal movement.**

3rd Trimester

- **The fetus increases in size.**
- **Circulatory and respiratory systems maturates.**
- **Increasing protein and calcium uptake from mother.**
- **Antibodies transfer from mother forming temporary immunity.**